A.L.F.I.E — Voice AI Enterprise Platform

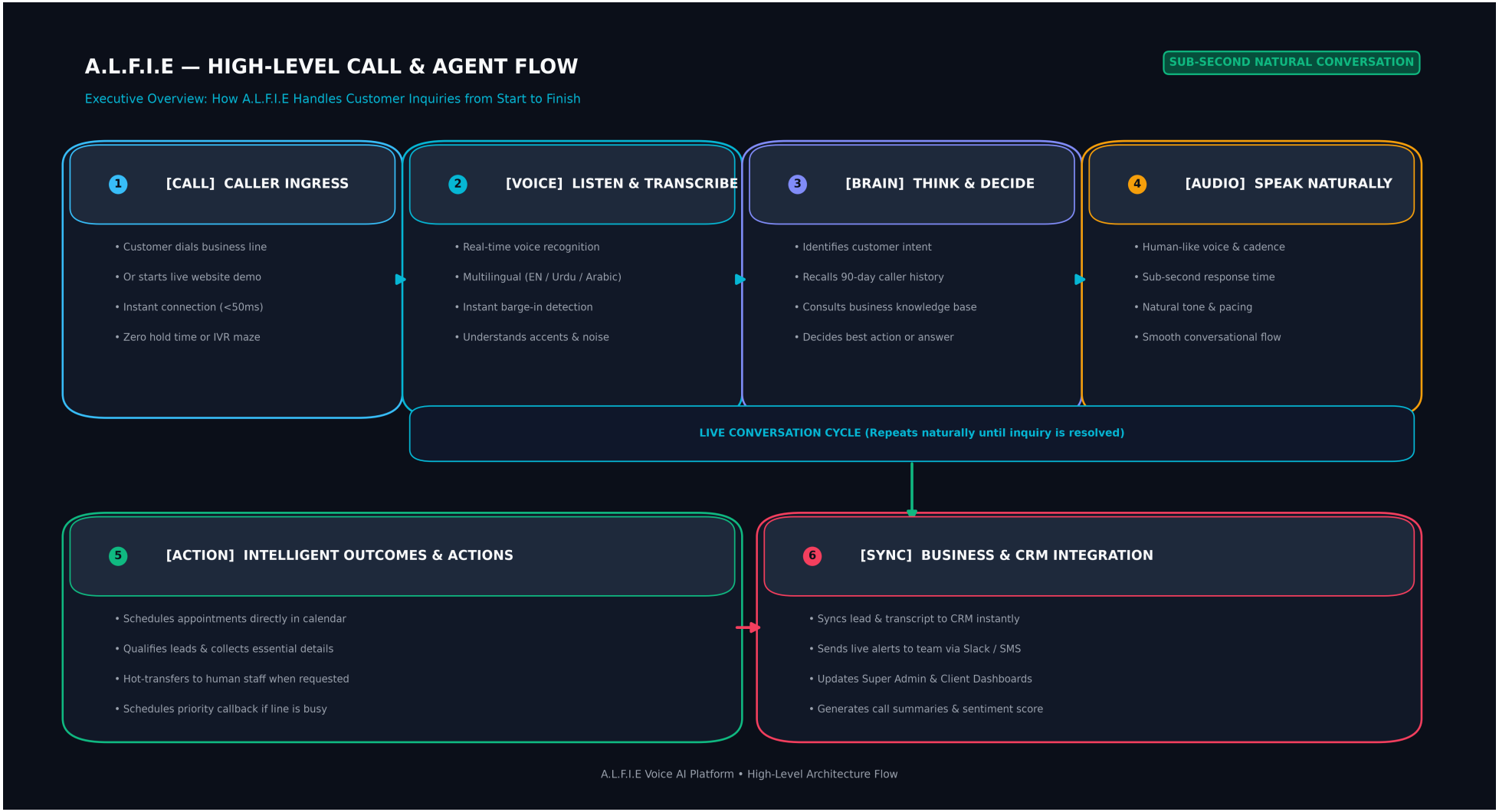
Architecture, Database E.R.D & End-to-End Call Lifecycle Flow Deck

A.L.F.I.E is an autonomous, sub-second latency Voice AI operating system engineered for enterprise inbound/outbound call operations, intelligent CRM appointment scheduling, lead qualification, and resilient hot-transfer routing across 5 distinct commercial verticals (Law Firms, Medical Clinics, Real Estate, Luxury Car Rentals, and Hospitality).

Layer	Technologies & Protocols	Key Responsibilities & SLAs
Telephony & Media	LiveKit WebRTC Cloud / Self-Hosted, SIP Trunks (Twilio / Telnyx), Opus/PCM Codecs	Inbound PSTN dispatch, audio frame streaming, sub-25ms jitter buffer
Speech-to-Text (STT)	Deepgram Nova-3, AssemblyAI Streaming, Silero VAD	Real-time audio transcription, turn detection, 180ms latency SLA
Reasoning Core (LLM)	Gemini 2.5, OpenAI GPT-4o-mini, Groq Llama-3.3, CachedOpenAILLM	Vertical persona instruction, intent routing, function calling, 320ms TTFT
Speech Synthesis (TTS)	Cartesia Sonic (Sonic-1 / Multilingual), ElevenLabs Turbo v2	Human-like cadence, streaming PCM chunking, 190ms TTFA
In-Memory Caches & State	Redis Stack 7.4+ (Bloom, TimeSeries, Vector Search, Streams, Hashes)	90-day caller context memory, semantic prompt cache, spam dedup, stream ingest
Persistence & Telemetry	PostgreSQL 16 (AsyncPG connection pool 2-20), Redis Streams, WebSockets	Multi-tenant call recordings, billing ledgers, QA metrics, real-time admin sync

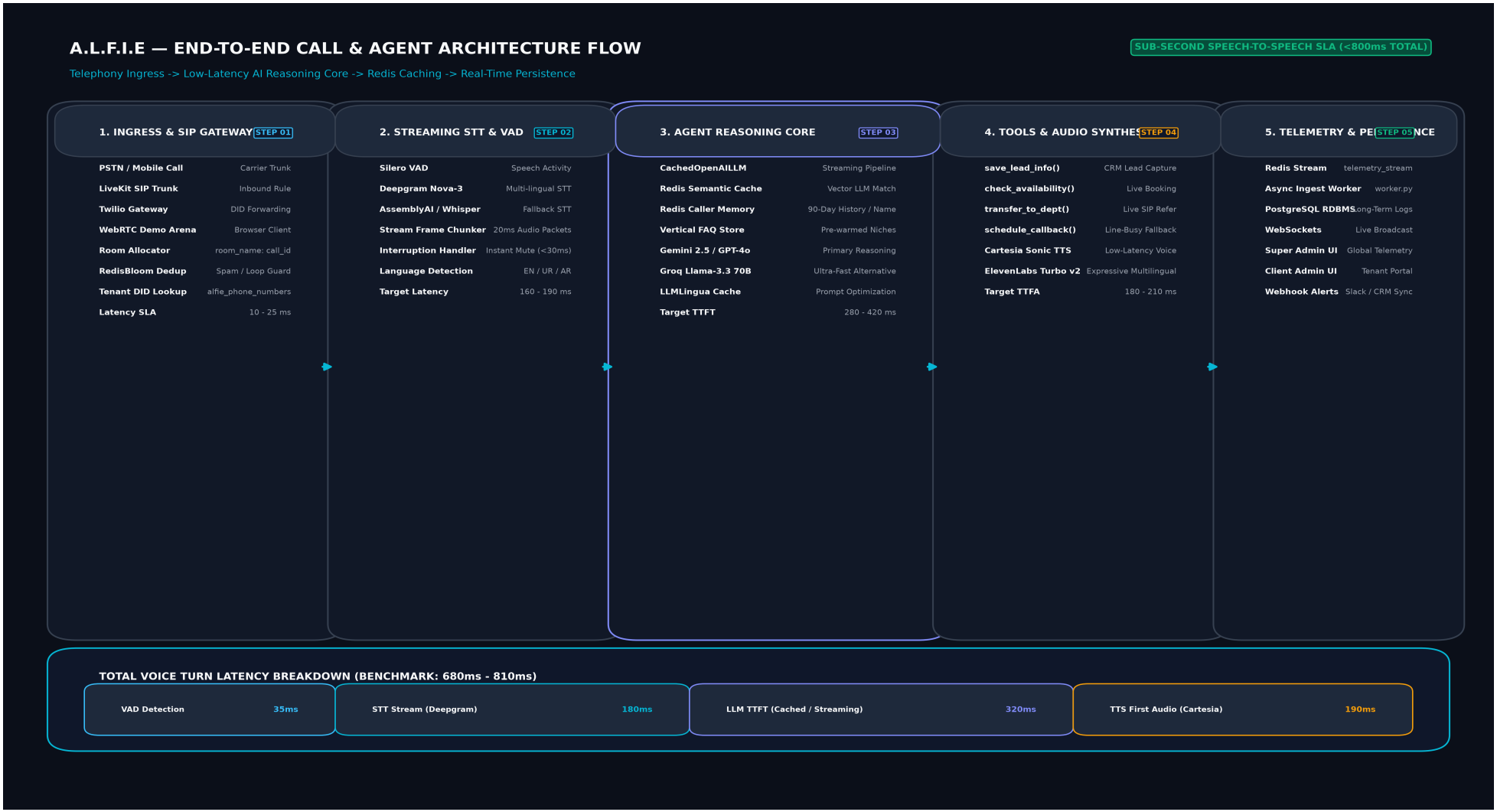
1. High-Level Call & Agent Flow (Executive)

Abstracted overview: Ingress -> Listen -> Think & Decide -> Speak -> Business Actions -> CRM Sync.



2. Deep Technical Call & Agent Lifecycle Flow

Sub-second speech-to-speech loop across LiveKit, Deepgram, LLM reasoning, Cartesia TTS, and Redis caching.



3. Relational Database Schema (E.R.D) & Memory Topology

PostgreSQL multi-tenant architecture, call recording ledgers, line-busy callbacks, and Redis Stack state.

